

Physical Chemistry (I)

Lecture 05

First Law of Thermodynamics



In the Last Class

- Internal energy
 - Kinetic molecular theory: monatomic perfect gas
 - Equipartition theorem: more general
- Real gas
 - Compression factor
 - Virial expansion
 - van der Waals equation

Last class

Internal Energy, U

“The total energy of a system”

For a perfect gas,

$$U = \sum_{i=1}^N \frac{1}{2} m v_i^2 = \frac{1}{2} M n \langle v^2 \rangle$$

$$U = \frac{3}{2} n R T$$

Equipartition Theorem

For a sample at thermal equilibrium,
the average value of each quadratic contribution
to the internal energy is $(1/2)k_B T$.

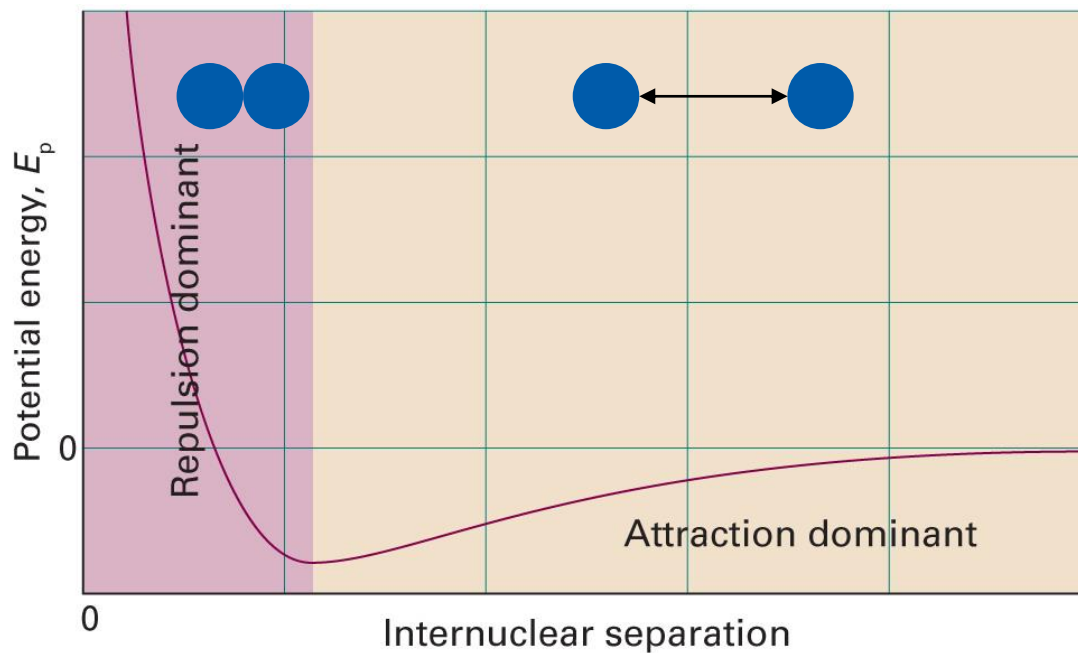
(Quadratic contribution: $E \propto p_x^2$ or $E \propto x^2$)

Translation + Rotation + Vibration

Last class

Real Gas

Now the molecules interact!



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1C.1)

Compression Factor Z

$$Z = V_m/V_m^0$$

Molar volume of a gas $V_m = V/n$

Molar volume of a perfect gas V_m^0

- $Z > 1$: $V_m > V_m^0 \rightarrow$ repulsion is dominant
- $Z \approx 1$: $V_m \approx V_m^0 \rightarrow$ perfect-gas-like behavior
- $Z < 1$: $V_m < V_m^0 \rightarrow$ attraction is dominant

Virial Expansion

$$Z(p) = 1 + B'p + C'p^2 + \dots$$

$$Z(V_m) = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots$$

- Second virial coefficient B
 - $B > 0 \rightarrow Z > 1 \rightarrow$ repulsion is dominant
 - $B = 0 \rightarrow Z \approx 1$ (perfect-gas-like) for a larger region
 - Boyle temperature T_B : temperature where $B = 0$
 - $B < 0 \rightarrow Z < 1 \rightarrow$ attraction is dominant

van der Waals Equation

- **Repulsion:** decrease of the total volume

$$V \rightarrow V - nb$$

$$p = \frac{nRT}{V - nb}$$

- **Attraction:** decrease of the pressure

$p \sim$ (frequency of collisions) $\times$ (force of each particle)

decreases as $n/V \uparrow$

decreases as $n/V \uparrow$

$$p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2}$$

van der Waals and Virial

- Virial expansion

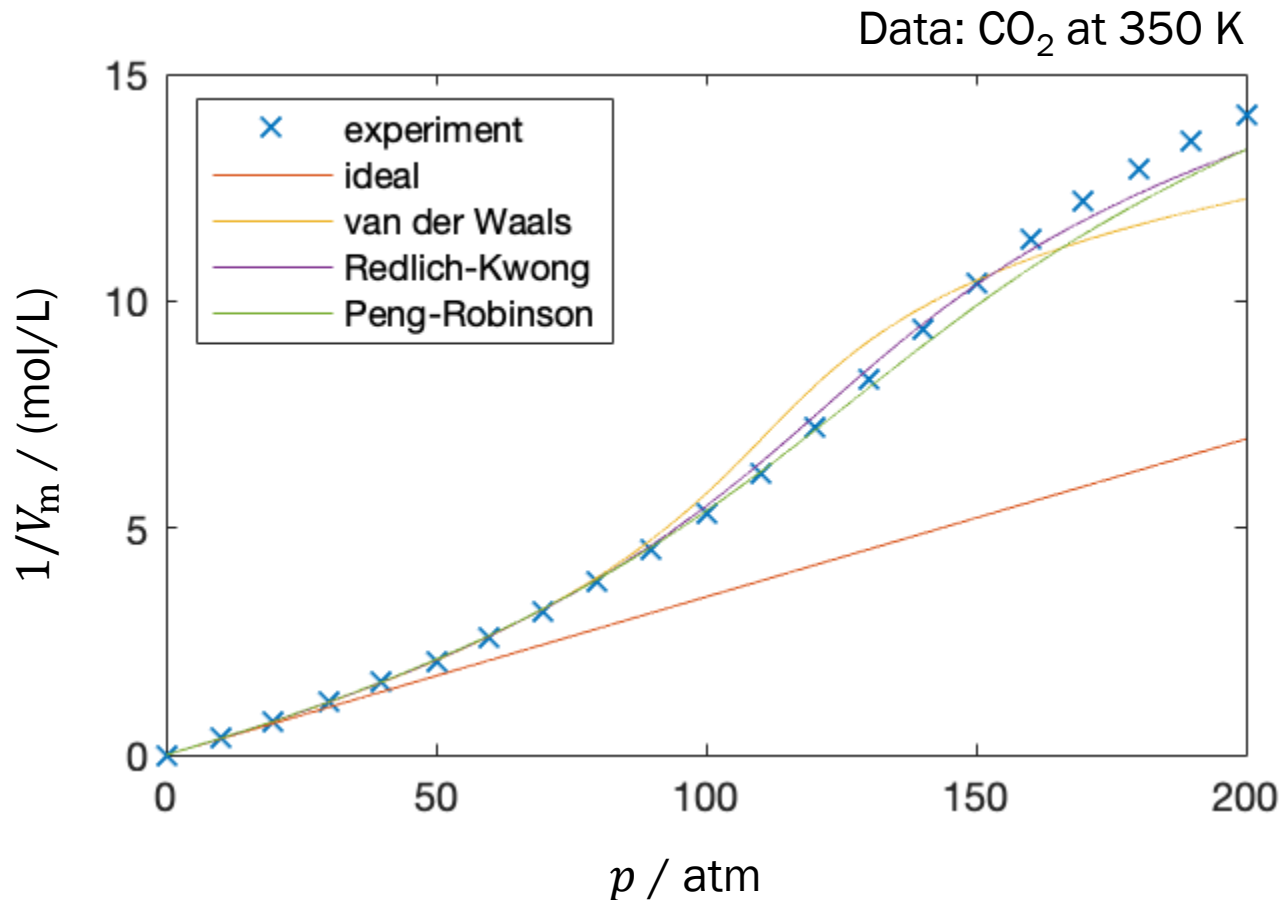
$$pV_m = RT \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right)$$

- van der Waals equation

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

$$pV_m = RT \left[1 + \left(b - \frac{a}{RT} \right) \frac{1}{V_m} + \frac{b^2}{V_m^2} + \dots \right]$$

Beyond van der Waals



3. The First Law



Chapter Overview

- First law of thermodynamics
 - Work and heat
 - Differential form of internal energy
- Enthalpy
 - Definition and properties of enthalpy
 - Differential form
 - Adiabatic change
- Thermochemistry



Internal Energy, U

“The total energy of a system”

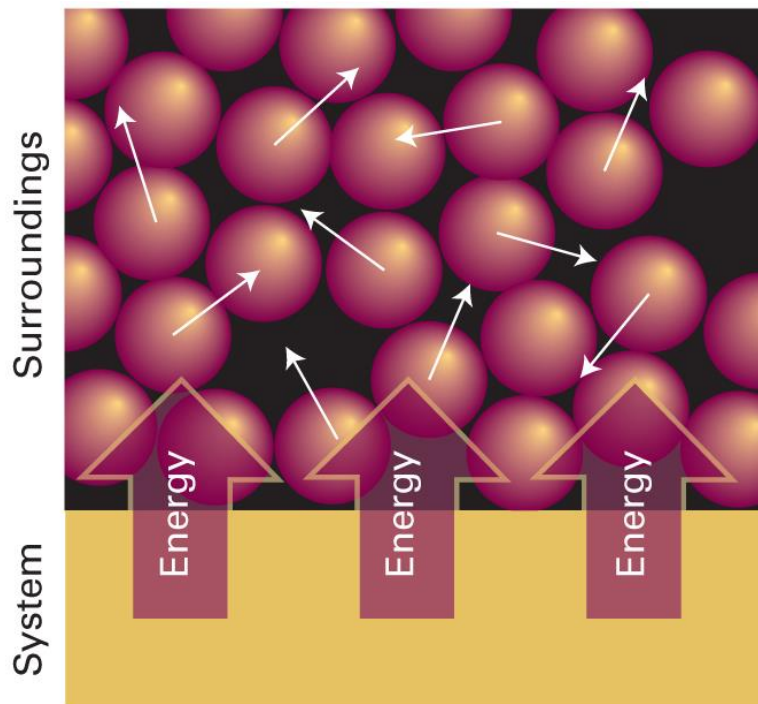
- Perfect gas (monatomic): $U = (3/2)nRT$
- Molar internal energy

$$U_m = U/n$$

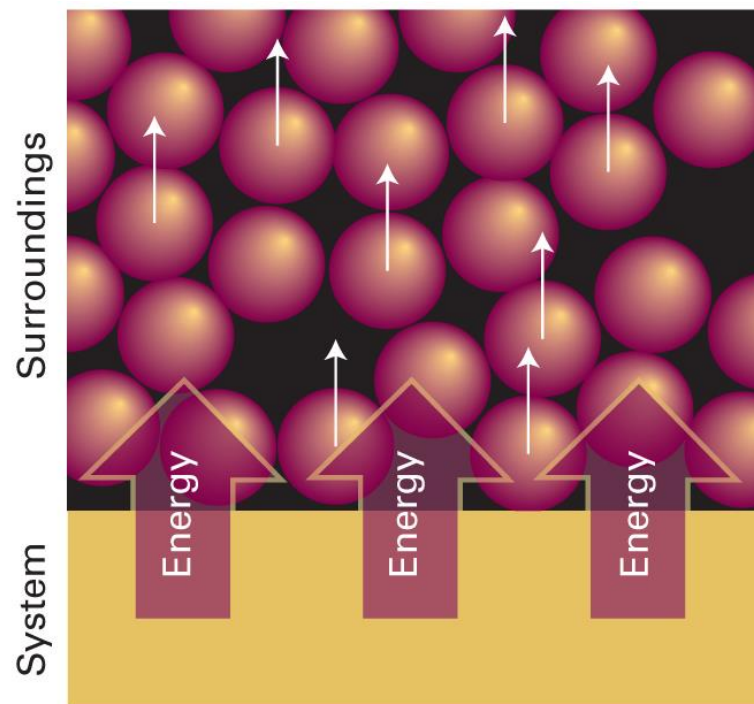
For a perfect gas, $U_m = (3/2)RT$

How can we change the internal energy?

Two Possible Ways



“Heat”



“Work”

First Law of Thermodynamics

“The internal energy of an isolated system is constant.”

- Heat and work are equivalent ways of changing the internal energy of a system

$$\Delta U = q + w$$

$$dU = \delta q + \delta w$$

↑ ↑ ↑
small change of U small q small w

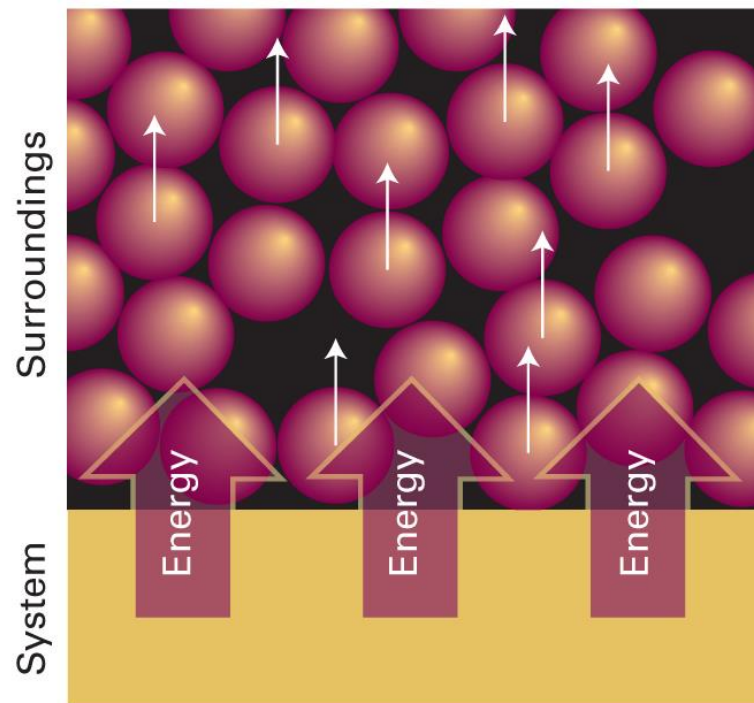
Work

$$\delta w = \vec{F} \cdot d\vec{s}$$

$\vec{F}$: “force” (intensive)

$d\vec{s}$: “displacement” (extensive)

- **Sign** indicates the direction of internal energy change



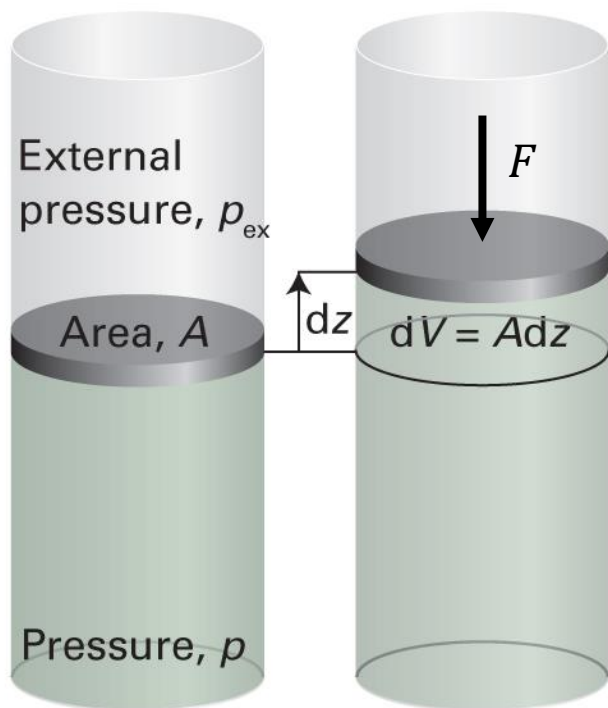
Several Types of Work

$$\delta w = (\text{intensive factor}) \times (\text{extensive factor})$$

Type of work	dw	Comments	Units [†]	
Expansion	$-p_{\text{ex}}dV$	p_{ex} is the external pressure dV is the change in volume	Pa m^3	} δw_{exp}
Surface expansion	$\gamma d\sigma$	γ is the surface tension $d\sigma$ is the change in area	N m^{-1} m^2	
Extension	$f dl$	f is the tension dl is the change in length	N m	} δw_{add}
Electrical	ϕdQ	ϕ is the electric potential dQ is the change in charge	V C	
	$Q d\phi$	$d\phi$ is the potential difference	V	
		Q is the charge transferred	C	



Expansion Work



$$\delta w = \vec{F} \cdot d\vec{s} = F_x dx + F_y dy + F_z dz$$

$$\text{As } dx = dy = 0,$$

$$\delta w = -|F|dz$$

$$\text{Since } p_{\text{ex}} = |F|/A,$$

$$\delta w = -p_{\text{ex}}Adz = -p_{\text{ex}}dV$$

Calculation of Expansion Work

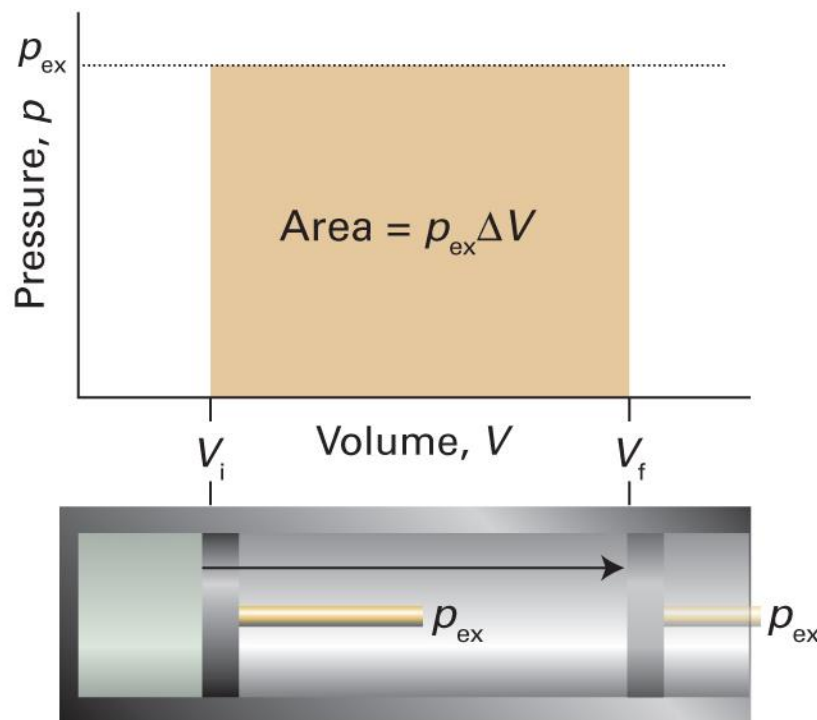
$$\delta w = -p_{\text{ex}} dV$$

- Total work for a change from **state i** to **state f**

$$w = \int_{\text{state } i}^{\text{state } f} \delta w = \int_{\text{state } i}^{\text{state } f} (-p_{\text{ex}} dV) = - \int_{V_i}^{V_f} p_{\text{ex}} dV$$

1. Expansion against constant external pressure
2. Reversible expansion

Constant External Pressure



$$w = - \int_{V_i}^{V_f} p_{\text{ex}} dV$$

$$w = -p_{\text{ex}} \int_{V_i}^{V_f} dV$$

$$w = -p_{\text{ex}}(V_f - V_i)$$

For $\Delta V = V_f - V_i$,

$$w = -p_{\text{ex}}\Delta V$$

Free Expansion

$$w = -p_{\text{ex}}\Delta V$$

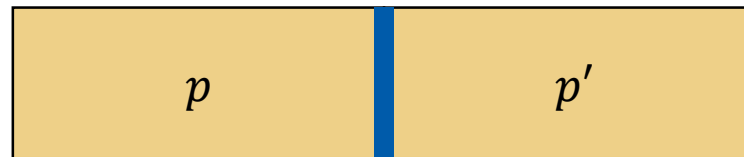
$$\text{If } p_{\text{ex}} = 0, w = 0$$

(regardless of the value of ΔV)

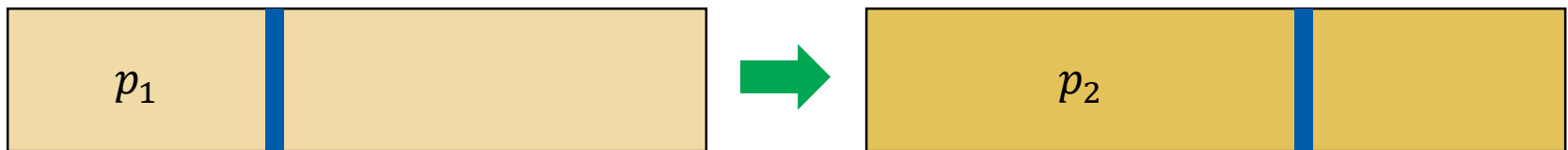
e.g. a gas expanding into a vacuum

Reversible Expansion

- Reversible change: a change that can be reversed by an **infinitesimal** modification of a variable
- The system is **at equilibrium** at any time during the change



$p = p'$ always



Reversible Expansion Work

$$w = - \int_{V_i}^{V_f} p_{\text{ex}} dV$$

For the reversible expansion, $p = p_{\text{ex}}$ always.

$$w = - \int_{V_i}^{V_f} p dV$$

Case of Perfect Gas

$$w = - \int_{V_i}^{V_f} p \, dV$$

For the reversible expansion of a perfect gas,

$$pV = nRT$$

$$p = nRT/V$$

Hence,

$$w = - \int_{V_i}^{V_f} p \, dV = - \int_{V_i}^{V_f} \frac{nRT}{V} dV = -nR \int_{V_i}^{V_f} \frac{T}{V} dV$$

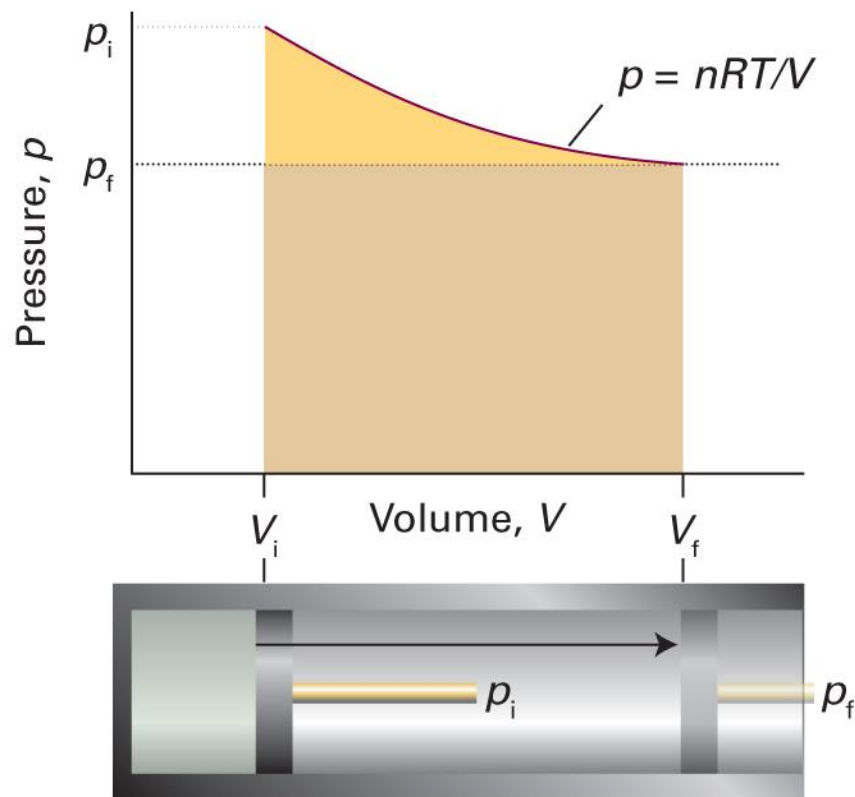
Perfect Gas, Constant T

For an **isothermal** reversible expansion,

$$w = -nR \int_{V_i}^{V_f} \frac{T}{V} dV = -nRT \int_{V_i}^{V_f} \frac{1}{V} dV$$

$$w = -nRT (\ln V_f - \ln V_i) = -nRT \ln \frac{V_f}{V_i}$$

Perfect Gas, Constant T



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 2A.7)

Internal Energy

$$dU = \delta q + \delta w \text{ (1st law)}$$

$$dU = \delta q + \delta w_{\text{exp}} + \delta w_{\text{add}}$$

$$dU = \delta q - p_{\text{ex}}dV + \delta w_{\text{add}}$$

For $\delta w_{\text{add}} = 0$,

$$dU = \delta q - p_{\text{ex}}dV$$

Heat

At constant volume,

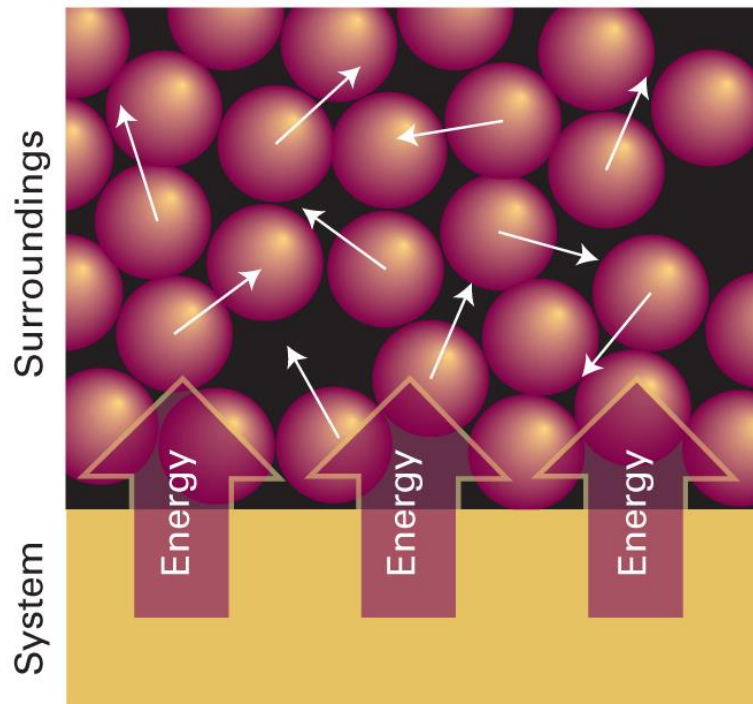
$$\delta q_V \propto dT$$

$$\delta q_V = C_V dT$$

At constant pressure,

$$\delta q_p \propto dT$$

$$\delta q_p = C_p dT$$



Heat Capacity

- Molar heat capacity $C_m = C/n$
- Specific heat capacity $C_s = C/m$
- Heat capacity depends on the constraint
 - C_V : heat capacity at constant volume ($C_{V,m}$, $C_{V,s}$)
 - C_p : heat capacity at constant pressure ($C_{p,m}$, $C_{p,s}$)

Calculation of Heat (Const V)

$$\delta q_V = C_V dT$$

- Total heat for a change from **state i** to **state f**

$$q_V = \int_{\text{state } i}^{\text{state } f} \delta q_V = \int_{\text{state } i}^{\text{state } f} (C_V dT) = \int_{T_i}^{T_f} C_V dT$$

Internal Energy (Const V)

$$dU = \delta q - p_{\text{ex}}dV$$

At constant volume, $dV = 0$ and $\delta q = C_V dT$

$$dU = C_V dT$$

The change in internal energy is

$$\Delta U = \int_{\text{state } i}^{\text{state } f} dU = \int_{\text{state } i}^{\text{state } f} (C_V dT) = \int_{T_i}^{T_f} C_V dT$$

At Constant V

$$\Delta U = q_V = \int_{T_i}^{T_f} C_V dT$$

If C_V is constant,

$$\Delta U = C_V \int_{T_i}^{T_f} dT = C_V(T_f - T_i) = C_V \Delta T$$

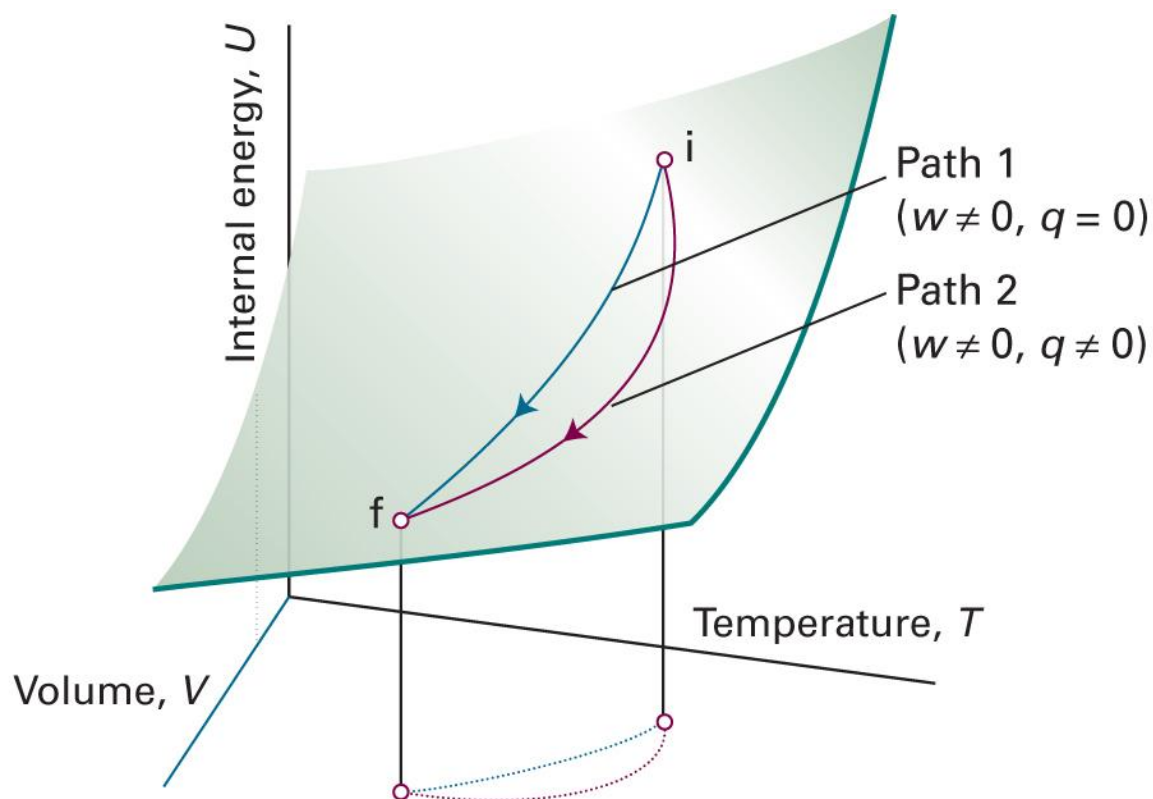
Now, get ready for some math...



State and Path Functions

- **State function:** a property that depends only on the current state of a system and is independent of its history
e.g. internal energy, pressure, volume, temperature
- **Path function:** a property with a value that depends on the path between two states
e.g. heat, work

State and Path Functions



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 2D.1)

Related Mathematics

- **Exact differential:** an infinitesimal quantity that, when integrated, gives a result that is independent of the path between the initial and final states.

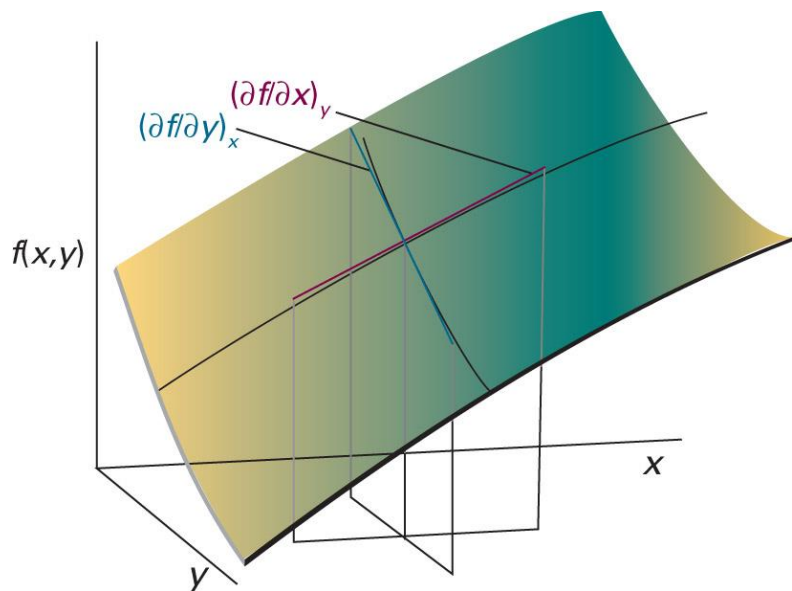
$$\Delta U = \int_{\text{state } i}^{\text{state } f} dU$$

- **Inexact differential:** an infinitesimal quantity that, when integrated, gives a result that depends on the path between the initial and final states.

$$q = \int_{\text{state } i, \text{ path}}^{\text{state } f} \delta q$$

Partial Derivatives

“the slope of the function with respect to one of the variables, all the other variables being held constant”



$$\left(\frac{\partial f}{\partial x}\right)_y = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

$$\left(\frac{\partial f}{\partial y}\right)_x = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$

Differential

If f is a function of x and y , the (total) differential df is

$$df = \left(\frac{\partial f}{\partial x} \right)_y dx + \left(\frac{\partial f}{\partial y} \right)_x dy$$

when df is an **exact differential**.

State Variables

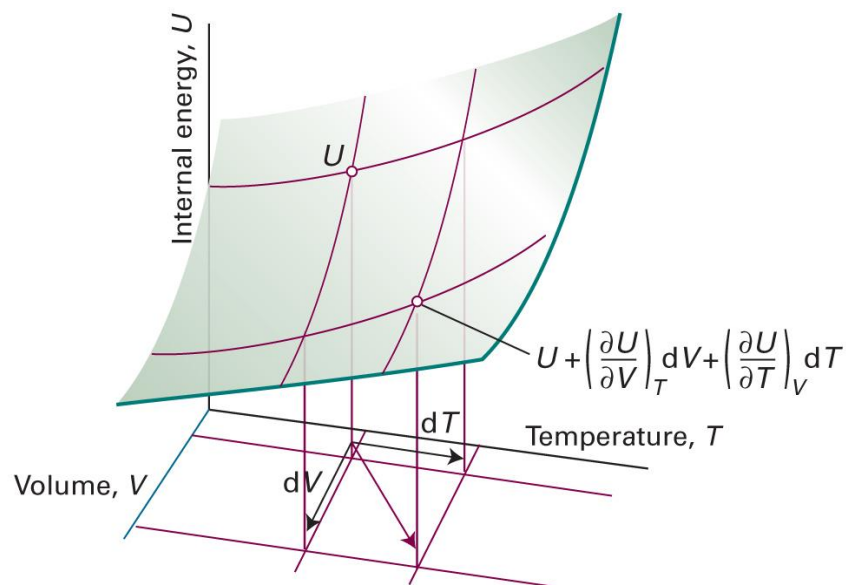
- A state is defined by p, V, n, T
- Typically n is fixed
- Equation of state: p is constrained by V and T
- We can just use V and T to define a state

$$U = U(V, T)$$

Changes in Internal Energy

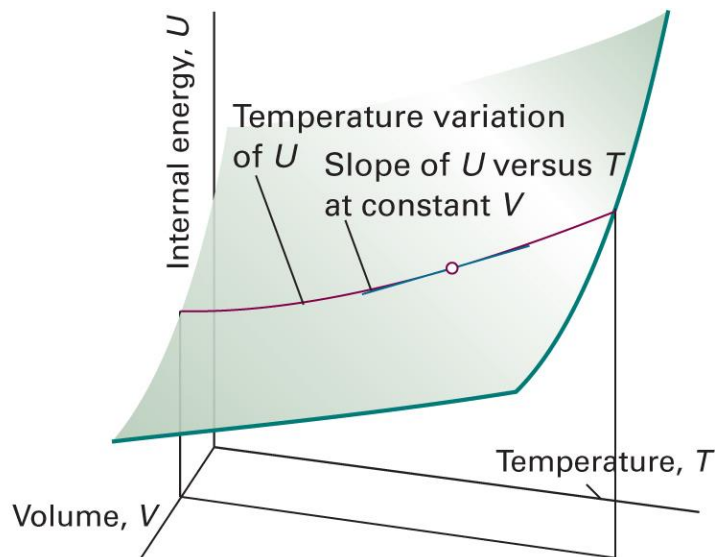
For $U = U(V, T)$,

$$dU = \left(\frac{\partial U}{\partial V} \right)_T dV + \left(\frac{\partial U}{\partial T} \right)_V dT$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 2D.2)

Meaning of $(\partial U / \partial T)_V$



$$dU = C_V dT \text{ (at constant } V\text{)}$$

$$\left(\frac{\partial U}{\partial T} \right)_V = C_V$$

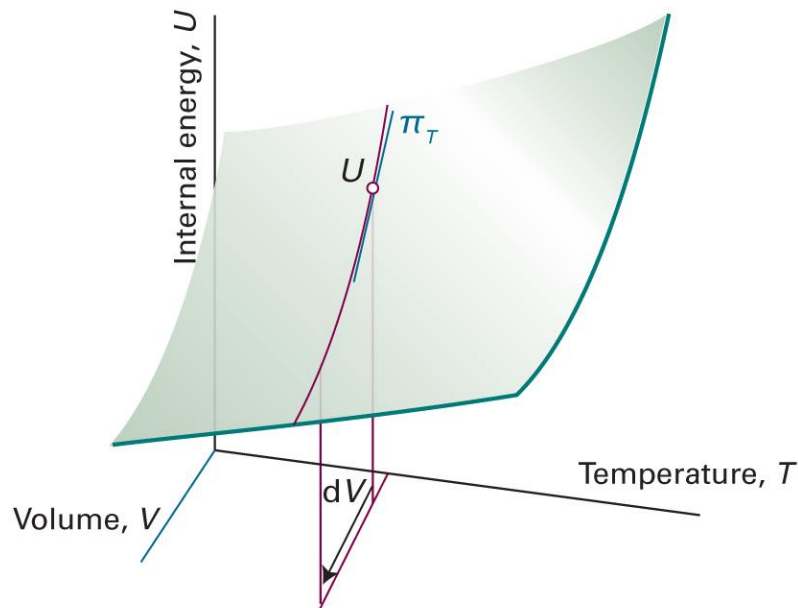
Not in Atkins

(Monatomic) Perfect Gas

$$U = \frac{3}{2}nRT$$

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = \left[\frac{\partial}{\partial T}\left(\frac{3}{2}nRT\right)\right]_V = \frac{3}{2}nR$$

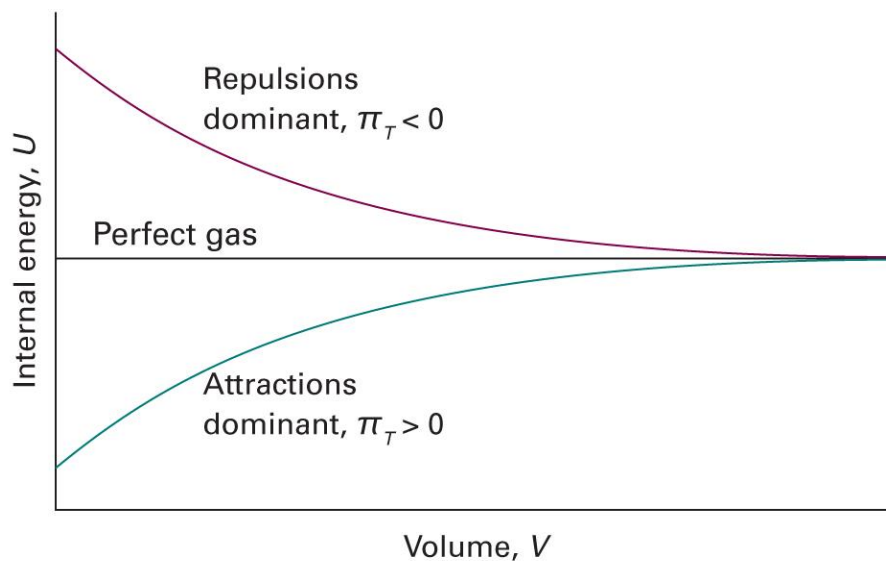
Meaning of $(\partial U / \partial V)_T$



variation of the internal energy of a substance
as its volume is changed at constant temperature

Internal Pressure

$$\pi_T = \left(\frac{\partial U}{\partial V} \right)_T$$



Use of Differentials

$$dU = \underbrace{\left(\frac{\partial U}{\partial V}\right)_T}_{\pi_T} dV + \underbrace{\left(\frac{\partial U}{\partial T}\right)_V}_{C_V} dT$$

$$dU = \pi_T dV + C_V dT$$

At constant pressure,

$$\left(\frac{\partial U}{\partial T}\right)_p = \pi_T \left(\frac{\partial V}{\partial T}\right)_p + C_V$$

Response Functions

- Expansion coefficient

$$\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p$$

- Isothermal compressibility

$$\kappa_T = -\frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_T$$

Some Data

	$\alpha / (10^{-4} \text{ K}^{-1})$	$\kappa_T / (10^{-6} \text{ bar}^{-1})$
<i>Liquids:</i>		
Benzene	12.4	90.9
Water	2.1	49.0
<i>Solids:</i>		
Diamond	0.030	0.185
Lead	0.861	2.18

General Expression

$$\left(\frac{\partial U}{\partial T}\right)_p = \pi_T \left(\frac{\partial V}{\partial T}\right)_p + C_V$$

Using the definition of the expansion coefficient α ,

$$\left(\frac{\partial U}{\partial T}\right)_p = \alpha \pi_T V + C_V$$

For a perfect gas, $\pi_T = 0$ and

$$\left(\frac{\partial U}{\partial T}\right)_p = C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

Example 2D.2

Derive an expression for the expansion coefficient of a perfect gas.

$$\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p$$

For a perfect gas, $V = nRT/p$

$$\left(\frac{\partial V}{\partial T} \right)_p = \left(\frac{\partial}{\partial T} \left(\frac{nRT}{p} \right) \right)_p = \frac{nR}{p}$$

Thus,

$$\alpha = \frac{1}{V} \frac{nR}{p} = \frac{nR}{nRT} = \frac{1}{T}$$

Summary for Chapter 3 (1)

First Law of Thermodynamics

$$dU = \delta q + \delta w$$

Work

$$\delta w = -p_{\text{ex}}dV + \delta w_{\text{add}}$$

- Constant- p change
- Reversible change

Heat

- Constant- V change

$$\delta q_V = C_V dT$$

$$C_V = (\partial U / \partial T)_V$$

Thermodynamics

Mathematics

State function

Exact differential

Path function

Inexact differential

Differential form of U

$$dU = \left(\frac{\partial U}{\partial V} \right)_T dV + \left(\frac{\partial U}{\partial T} \right)_V dT$$

- Internal pressure π_T
- Response functions α , κ_T